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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21BT502/20BT502 – ENZYME TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Write about major classification of enzymes with examples.	10	L*2	1	1,2
b)	Enzymes do not change the free energy of reaction rather an enzyme lowers the energy of activation. Discuss with an illustration.	10	L2	1	1,2
2. a)	Describe the role of activation energy in enzyme catalyzed reactions.	10	L2	1	1,2
b)	Comment on Lock and key hypothesis of enzyme action.	10	L2	1	1,2
Unit – II					
3. a)	What are the various factors affecting enzyme activity? Discuss.	10	L2	2	1,2
b)	How classification of chemical reactions by kinetic order is done? Explain.	10	L2	3	1,2
4. a)	Optimum pH is required for maximum enzyme activity. Justify.	10	L2	3	1,2
b)	Discuss about Noncompetitive inhibition with a suitable example.	10	L2	2	1,2
5. a)	Comment on enzyme kinetics with reference to pseudo-order and first order reactions.	10	L2	2	1,2
b)	Explain metal ion catalysis of enzymatic reactions with suitable example.	10	L2	3	1,2
Unit – III					
6. a)	Describe high throughput enzyme screening.	10	L2	4	1,2
b)	With suitable example explain extraction and purification of enzymes.	10	L2	4	1,2
7. a)	Discuss about industrial applications of immobilized enzymes.	10	L2	4	1,2
b)	Narrate the molecular determination and characterization of enzymes.	10	L1	5	1,2
8. a)	Give a brief account on criteria of purity.	10	L2	5	1,2
b)	Comment on bioreactors using immobilized enzymes.	10	L1	4	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
